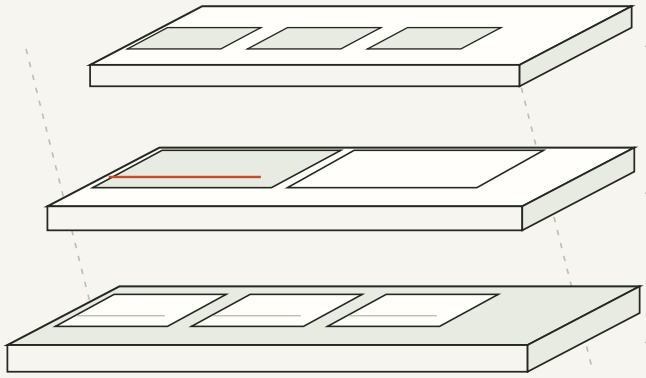
SYSTEM OVERVIEW

# lattices

A programmable workspace for macOS. Arrange windows, keep project sessions, find context, and give agents a local control surface.

FIG. A / FUNCTIONAL ASSEMBLY

SCHEMATIC / NOT TO SCALE



**01** **Workspace system**
Projects / sessions / layers

**02** **Desktop control**
Windows / displays / input

**03** **Agent system**
Observe / act / capture

ONE DESKTOP / SHARED CONTEXT / MULTIPLE INTERFACES

CAPABILITY MAP

## 01 Workspace system

SEE SHEET 02

Project discovery; declarative tmux panes; launch, attach and repair; configured tab groups; live tab stacks; layers; Overview; search and terminal maps.

## 02 Desktop control

SEE SHEET 03

Focus and place windows; tile and distribute; move between displays and Spaces; arrange in place; use gestures, hotkeys, command bar, OCR and voice input.

## 03 Agent system

SEE SHEET 04

Local API and SDK; browser MCP; computer actions; screenshots and recordings; run artifacts; overlay HUDs; workspace assistant; paired mobile controls.

COMPANION PRODUCTS

**Action**
Computer use, browser work, guided capture and recording.

**Blink**
Floating notes, named desks and offline mobile reading.

**Voice / Speech**
Queued readouts, voice selection and a playback HUD.

macOS 26+ / NATIVE APP + CLI / tmux OPTIONAL / COMPANIONS INSTALL SEPARATELY

| DRAWING | ISSUE / BASIS | SOURCE SNAPSHOT | SHEET / SERIES |
|---|---|---|---|
| **LAT-FS-00** | A / Working source | 02 Oct 2026 | **01 / 04** |

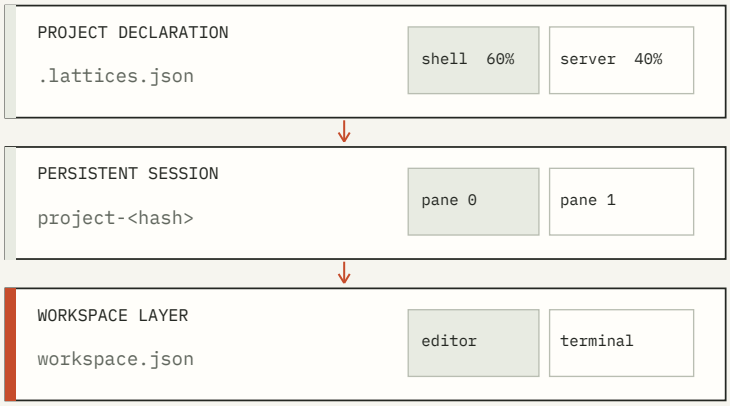
SUBSYSTEM 01 / ORGANIZE

# Workspace system

Keep a project's working context in named sessions, tab groups and layers. Inspect the desktop without losing the distinction between saved and live state.

FIG. B / CONTEXT ASSEMBLY

SCHEMATIC / NOT TO SCALE




**01 Declare**

Panes, commands, placement

**02 Resume**

Attach, sync, restart

**03 Compose**

Match, focus, tile, reveal

CONFIGURATION -> SESSION STATE -> DESKTOP MEMBERSHIP

FUNCTIONAL SPECIFICATION

### 01 Projects & development

Discover configs, choose among six terminals and launch sessions. Detect dev/build commands for Node, Swift, Rust, Go and Make.

### 02 Session lifecycle

Declare named panes and sizes. List, detach or kill sessions; sync missing panes; restart commands; ensure or prefill on reattach.

### 03 Tab groups

Launch configured project groups and cycle tabs. Build temporary stacks from live windows, then dissolve them when finished.

### 04 Layer rules

Match projects, apps or specific windows. Create, rename, delete, capture and edit layers; add or remove membership.

### 05 Layer switching

Activate, launch and arrange a layer. Park or hide other windows; Show All restores them. Eight direct slots surround a center status key.

### 06 Workspace views

Overview, Studio and HUD maps expose windows and layers. Inspect saved, last-known and live positions; preview, focus and arrange.

### 07 Find context

Search titles, apps, session tags and OCR. Deep search adds live terminal titles, working directories, commands and tmux context.

### 08 Maps & inventory

Read the desktop as an ASCII map or JSON. Inspect managed sessions, terminal instances, processes, display frames and window IDs.

STATE CONTRACT

Project config defines panes. workspace.json defines layers and groups. Live membership resolves against current windows; CLI map reads do not move them.

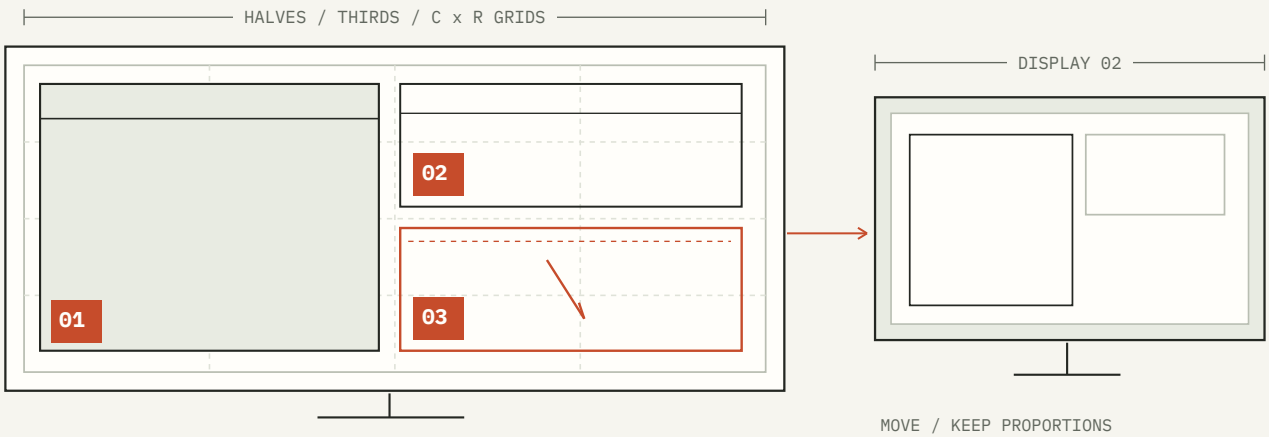
SUBSYSTEM 02 / ARRANGE

# Desktop control

Shape the workspace through direct manipulation, keyboard chords, gestures and commands, with previews and explicit restore paths where supported.

FIG. C / DISPLAY PLAN + WINDOW PLACEMENT

SCHEMATIC / NOT TO SCALE




FUNCTIONAL SPECIFICATION

### 01 Place & distribute

Halves, quarters, thirds, corners, maximize and custom grids. Fill open cells, distribute selected windows or tile an app family.

### 02 Displays & Spaces

Focus a target window, move it to another monitor or Space, preserve relative geometry, and switch desktops with instant transitions.

### 03 In-place tools

Select real windows; Fill, Grid and Swap act live. Place stages a change. Keep accepts it; Restore returns captured frames and stacking order.

### 04 Navigation surfaces

Command bar, Omni search, window hints, Hyperspace survey, spatial lens, screen map, layer bezel and configurable HUD presets.

### 05 Keyboard & pointer

Edit shortcuts, Caps Lock Hyper/Escape remaps and per-device gesture rules. Use the aim HUD, drag snap, cursor finder and center summon.

### 06 Screen text

AX-first quick scans, deeper Vision OCR, indexed search, recent scans and per-window history. Retention and scan cadence are configurable.

### 07 Voice & hands-off

Push-to-talk and spoken workspace intents; natural-language command preview and advisor. Hands-off mode accepts continued spoken turns.

### 08 Focus & preferences

Spotlight and restore a window with Focus Mode. Configure launch, input, AI and companions; inspect diagnostics, update and relaunch the app.

OPERATING CONDITIONS

Accessibility enables window and input control. Screen Recording enables screen capture/OCR. Voice input depends on a voice-enabled build and configured services.

DRAWING
**LAT-FS-02**

ISSUE / BASIS
A / Working source

SOURCE SNAPSHOT
02 Oct 2026

SHEET / SERIES
**03 / 04**

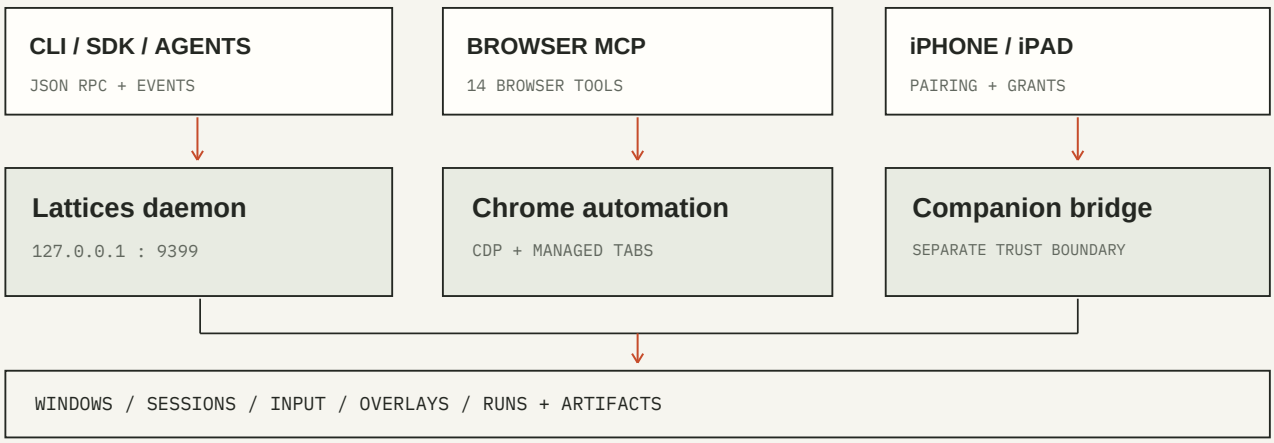
SUBSYSTEM 03 / AUTOMATE

# Agent system

Expose workspace state, actions and evidence to scripts, agents and paired devices. Keep browser tools and companion products explicit in the topology.

FIG. D / CONTROL + EVIDENCE TOPOLOGY

SCHEMATIC / NOT TO SCALE




FUNCTIONAL SPECIFICATION

### 01 Native API & SDK

Local WebSocket RPC, a runtime schema, event subscriptions and TypeScript facades expose workspace, desktop, input and system operations.

### 02 Observe & inspect

Read windows, Spaces, layers, sessions, terminal output and process trees. Resolve targets; inspect AX elements, OCR and screenshots.

### 03 Computer actions

Launch, focus, click, type, set values, press keys, scroll and drag. Stage actions, inspect receipts, verify outcomes and undo eligible placements.

### 04 Capture & runs

Capture windows, regions or displays; record a target or command. Save runs, steps, logs and artifacts; crop/zoom images and apply verification.

### 05 Browser interface

Browser MCP controls isolated Chrome identities: navigate, inspect, click, fill, resize, capture and read console logs. Native browser RPC is separate.

### 06 Assistant & overlays

Workspace chat selects an available agent backend. Publish transient overlays, interactive actors and web HUDs; move them or control visibility.

### 07 Paired mobile control

A separate bridge pairs iPhone/iPad devices with grants and revocation. Deck controls, desktop previews, keyboard and trackpad are capability-gated.

### 08 Product extensions

Separate apps: Action for computer use; Blink for notes and offline mobile reading; Voice for speech queues, voice selection and playback controls.

INTERFACE CONTRACT

The native daemon is loopback-only with no auth by design. Browser MCP is a distinct toolset. Companion access requires pairing; voice input is hosted on :9398.

DRAWING
**LAT-FS-03**

ISSUE / BASIS
A / Working source

SOURCE SNAPSHOT
02 Oct 2026

SHEET / SERIES
**04 / 04**